

PATIENT MULTI-OMICS GENOMIC HEALTH SUMMARY

Patient Submitter ID: TCGA-BH-A0B2 | Tumor: Breast Invasive Carcinoma | Stage: Stage IIA

TCGA Project:	TCGA-BRCA	cBioPortal Study:	brca_tcg_pan_can_atlas_2018
Gender / Age:	female / 58 yrs	Vital Status / OS:	Alive (64.2 mo)
Hybrid ML Risk:	78.5% (CRITICAL RISK)	VQC / Classical:	74.5% / 81.2%

FOUNDATIONAL 6-DATABASE MULTI-OMICS MUTATION PROFILE (GDC, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl)

Gene	Locus (GRCh38)	Transcript	Protein Change	ClinVar Status	COSMIC ID	gnomAD Pop AF	VAF %
BRCA1	chr17:17q21.31	ENST00000357654	p.E23fs / 185delAG Frameshift Truncation	Pathogenic (HRD Deficient) VCV000017659	COSV51862901 FATHMM: 0.99 Pathogenic	0.000021	48.6%
TP53	chr17:17p13.1	ENST00000269305	p.R175H Missense Mutation	Pathogenic / Li-Fraumeni VCV000012374	COSV52669389 FATHMM: 0.99 Pathogenic	0.000016	42.8%

QUANTUM VARIATIONAL ROTATION STATE ANGLES (θ VQC / R_Y / CNOT)

Gene Symbol	Canonical Ensembl ID	VAF Hotspot Frequency	Variational Angle θ (Radians)	Qubit Hilbert Space Projection
BRCA1	ENSG00000012048	48.6%	$\theta = 1.8420 \text{ rad (105.5}^\circ\text{)}$	$ \psi\rangle = \cos(0.921) 0\rangle + \sin(0.921) 1\rangle$
TP53	ENSG00000141510	42.8%	$\theta = 2.1480 \text{ rad (123.1}^\circ\text{)}$	$ \psi\rangle = \cos(1.074) 0\rangle + \sin(1.074) 1\rangle$

PRECISION ONCOLOGY CLINICAL ACTION & PROTOCOL RECOMMENDATION

Recommended Therapeutic Pathway:

Multidisciplinary tumor board review, BRCA1 germline testing confirmation, and PARP inhibitor eligibility evaluation.

Genomic Biomarker Summary: Identified somatic hotspot mutations in BRCA1 (p.E23fs / 185delAG), TP53 (p.R175H). Confirmed pathological classification across ClinVar and COSMIC registries with extreme somatic rarity in gnomAD exomes.

NOTICE & RESEARCH DISCLAIMER: This report is generated by the Blazeфинix Quantum-Classical Multi-Omics Research Platform using open-access academic cancer genomics datasets (NCI GDC TCGA, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl). It is calibrated strictly for computational research, translational bioinformatics, and clinical decision support benchmarking, not for standalone autonomous medical diagnosis.